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ELECTRONIC PRINCIPLES AND CIRCUITS (BEC303)

MODULE – 1 TRANSISTOR BIASING, BJT AC MODELS, VOLTAGE AMPLIFIERS & CC AMPLIFIERS

Topics Covered:

- Voltage Divider Bias • VDB Analysis • VDB Load Line & Q Point
 - Two Supply Emitter Bias • Other Types of Bias
- BJT AC Models • Small Signal Operation • AC Beta • r_e'
 - T Model • π Model • H Parameters • Voltage Gain
 - Loading Effect • CC Amplifier • Output Impedance

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PART A: TRANSISTOR BIASING

1. Voltage Divider Bias (VDB)

Voltage Divider Bias (VDB) is the most widely used transistor biasing method in practical amplifier circuits. The circuit gets its name from the **voltage divider network** formed by resistors R_1 and R_2 connected between V_{CC} and ground, which establishes the base voltage.

The key advantage of VDB is its excellent Q-point stability. Since the base voltage V_{BB} is derived from the voltage divider (independent of transistor parameters like β), the Q point remains stable despite transistor replacement or temperature changes.

Circuit Description:

- R_1 and R_2 form a voltage divider between V_{CC} and ground
- The junction of R_1 and R_2 connects to the transistor base
- R_C (collector resistor) connects between V_{CC} and collector
- R_E (emitter resistor) connects between emitter and ground
- In a well-designed VDB circuit, the base current I_B is negligible compared to the voltage divider current, making the base voltage essentially independent of β

Transistor Biasing

Voltage Divider Biasing

The figure 8.1a shows the voltage divider biasing circuit. Notice that the base circuit contains a voltage divider (R_1 and R_2) because of this circuit is called voltage divider bias (VDB).

VDB Analysis

In any well designed VDB circuit the base current is very negligible compared to current through voltage divider circuit and hence equivalent circuit at base is shown in figure 8.1b.

To analyse VDB circuit we need follow the steps given below.

1. Calculate the base voltage V_{BB} out of voltage divider bias which is given by

$$V_{BB} = \frac{R_2}{R_1 + R_2} V_{CC}$$

With V_{BB} as a dc source the equivalent circuit for figure 8.1a is given in figure 8.1c.

2. Subtract 0.7V to get the emitter voltage.
 $V_E = V_{BB} - V_{BE}$
3. Divide the V_E by R_E to get emitter current.
 $I_E = V_E / R_E$
4. Assume the collector current is equal to emitter current.
 $I_C \approx I_E$
5. Calculate collector voltage using following equation.
 $V_C = V_{CC} - I_C R_C$
6. Then calculate collector emitter voltage (V_{CE}) by subtracting emitter voltage from collector voltage.
 $V_{CE} = V_C - V_E$

Figure 8-1 Voltage-divider bias.
(a) Circuit; (b) voltage divider;
(c) simplified circuit.

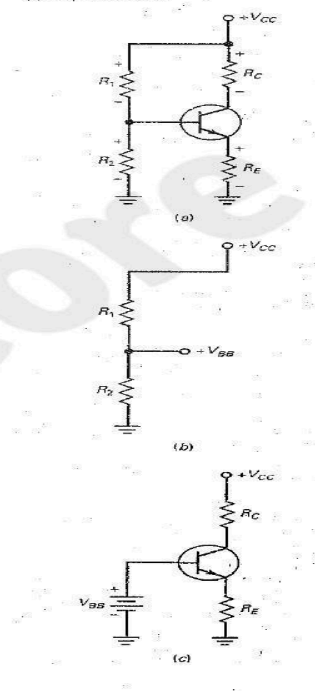


Figure: Voltage Divider Bias circuit (a) Full circuit (b) Simplified equivalent (c) Thevenin equivalent

1.1 VDB Analysis – Step-by-Step

In any well-designed VDB circuit, the base current is very negligible compared to current through the voltage divider circuit. Hence the Thevenin equivalent circuit at the base simplifies the analysis considerably.

Step-by-Step VDB Analysis:

1. **Calculate base voltage V_{BB} :** The voltage divider produces an unloaded output voltage:

$$V_{BB} = (R_2 / (R_1 + R_2)) \times V_{CC}$$

2. **Calculate emitter voltage V_E :** Subtract the base-emitter diode drop (0.7V):

$$V_E = V_{BB} - V_{BE} = V_{BB} - 0.7 \text{ V}$$

3. **Calculate emitter current I_E :** Divide emitter voltage by emitter resistor:

$$I_E = V_E / R_E$$

4. **Approximate collector current I_C :** Since alpha is very close to 1:

$$I_C \approx I_E$$

5. **Calculate collector voltage V_C :** Using Kirchhoff's voltage law:

$$V_C = V_{CC} - I_C \times R_C$$

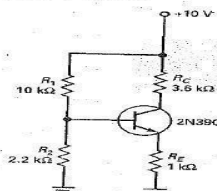
6. **Calculate V_{CE} :** Collector-emitter voltage:

$$V_{CE} = V_C - V_E$$

Solved Example 8-1: VDB circuit with $V_{CC} = +10\text{V}$, $R_1 = 10\text{k}\Omega$, $R_2 = 2.2\text{k}\Omega$, $R_C = 3.6\text{k}\Omega$, $R_E = 1\text{k}\Omega$ (2N3904 transistor)

Example 8-1

Figure 8-2 Example.



What is the collector-emitter voltage in Fig. 8-2?

SOLUTION The voltage divider produces an unloaded output voltage of:

$$V_{BB} = \frac{2.2 \text{ k}\Omega}{10 \text{ k}\Omega + 2.2 \text{ k}\Omega} 10 \text{ V} = 1.8 \text{ V}$$

Subtract 0.7 V from this to get:

$$V_E = 1.8 \text{ V} - 0.7 \text{ V} = 1.1 \text{ V}$$

The emitter current is:

$$I_E = \frac{1.1 \text{ V}}{1 \text{ k}\Omega} = 1.1 \text{ mA}$$

Since the collector current almost equals the emitter current, we can calculate the collector-to-ground voltage like this:

$$V_C = 10 \text{ V} - (1.1 \text{ mA})(3.6 \text{ k}\Omega) = 6.04 \text{ V}$$

The collector-emitter voltage is:

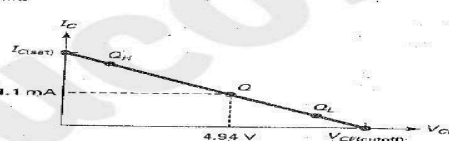
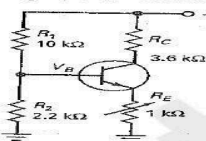
$$V_{CE} = 6.04 - 1.1 \text{ V} = 4.94 \text{ V}$$

Here is an important solution:

VDB load line and Q point

The Q point

Figure 8-6 Calculating the Q point.



From the example 8.1 the Q point is calculated which is at the collector current of 1.1 mA and a collector emitter voltage of 4.94 V. These values are plotted to get the Q point shown in Figure 8.6. Since the voltage divider bias is derived from emitter bias, one way to move the Q point of figure 8.6 is by varying the emitter resistor.

For instance, if the emitter resistance is changed to 2.2 kΩ, the collector current decreases to:

$$I_E = \frac{1.1 \text{ V}}{2.2 \text{ k}\Omega} = 0.5 \text{ mA}$$

The voltages change as follows:

$$V_C = 10 \text{ V} - (0.5 \text{ mA})(3.6 \text{ k}\Omega) = 8.2 \text{ V}$$

and

$$V_{CE} = 8.2 \text{ V} - 1.1 \text{ V} = 7.1 \text{ V}$$

Therefore, the new Q point will be Q_E and will have coordinates of 0.5 mA and 7.1 V.

On the other hand, if we decrease the emitter resistance to 510 Ω, the emitter current increases to:

$$I_E = \frac{1.1 \text{ V}}{510 \Omega} = 2.15 \text{ mA}$$

Figure: Example 8-1 solved – VDB circuit with Q point calculation

Solution:

$$V_{BB} = (2.2 \text{ k}\Omega / (10 \text{ k}\Omega + 2.2 \text{ k}\Omega)) \times 10 \text{ V} = 1.8 \text{ V}$$

$$V_E = 1.8 \text{ V} - 0.7 \text{ V} = 1.1 \text{ V}$$

$$I_E = 1.1 \text{ V} / 1 \text{ k}\Omega = 1.1 \text{ mA}$$

$$I_C \approx I_E = 1.1 \text{ mA}$$

$$V_C = 10 \text{ V} - (1.1 \text{ mA})(3.6 \text{ k}\Omega) = 6.04 \text{ V}$$

$$V_{CE} = 6.04 \text{ V} - 1.1 \text{ V} = 4.94 \text{ V}$$

Therefore, the Q point is at $I_C = 1.1 \text{ mA}$ and $V_{CE} = 4.94 \text{ V}$.

✚ **NOTE:** The VDB Q point is very stable because V_{BB} is set purely by R_1 , R_2 , and V_{CC} — it does not depend on transistor beta (β). This makes VDB the most reliable biasing method.

1.2 VDB Load Line and Q Point

The DC load line is a straight line on the I_C - V_{CE} characteristics plane that represents all possible operating points (I_C , V_{CE}) for a given circuit. The Q point (Quiescent Point) is the specific DC operating point at which the transistor operates in the absence of an AC signal.

Load Line Endpoints:

- At $I_C = 0$ (cutoff): $V_{CE} = V_{CC}$ — this gives the right endpoint on the x-axis
- At $V_{CE} = 0$ (saturation): $I_{C(sat)} = V_{CC}/(R_C + R_E)$ — this gives the top endpoint on the y-axis
- The Q point lies somewhere on this load line, ideally near the middle for maximum undistorted swing

Effect of Emitter Resistor on Q Point:

Since VDB is derived from emitter bias, one way to move the Q point is by varying the emitter resistor R_E .

- Increasing R_E : moves Q point DOWN the load line (lower I_C , higher V_{CE}) → Q point shifts to QL
- Decreasing R_E : moves Q point UP the load line (higher I_C , lower V_{CE}) → Q point shifts to QH

Example – Varying R_E :

If R_E changes from $1 \text{ k}\Omega$ to $2.2 \text{ k}\Omega$ ($V_{BB} = 1.1 \text{ V}$ unchanged):

$$I_E = 1.1 \text{ V} / 2.2 \text{ k}\Omega = 0.5 \text{ mA} \rightarrow V_C = 10 \text{ V} - (0.5 \text{ mA})(3.6 \text{ k}\Omega) = 8.2 \text{ V} \rightarrow V_{CE} = 8.2 \text{ V} - 1.1 \text{ V} = 7.1 \text{ V}$$

New Q point: $I_C = 0.5 \text{ mA}$, $V_{CE} = 7.1 \text{ V}$ (moves to QL — lower on load line)

If R_E decreases to 510Ω :

$$I_E = 1.1 \text{ V} / 510 \Omega = 2.15 \text{ mA} \rightarrow V_C = 10 \text{ V} - (2.15 \text{ mA})(3.6 \text{ k}\Omega) = 2.26 \text{ V} \rightarrow V_{CE} = 2.26 \text{ V} - 1.1 \text{ V} = 1.16 \text{ V}$$

New Q point: $I_C = 2.15 \text{ mA}$, $V_{CE} = 1.16 \text{ V}$ (moves to QH — higher on load line)

2. Two Supply Emitter Bias (TSEB)

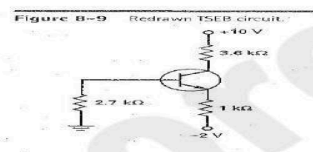
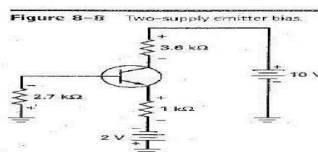
Two Supply Emitter Bias (TSEB) uses **two separate power supplies** — a positive supply +VCC and a negative supply -VEE — to bias the transistor. The negative supply (-VEE) forward biases the emitter diode, while the positive supply (+VCC) reverse biases the collector diode.

This circuit is derived from the emitter bias configuration. Because of the dual supply, the base voltage is ideally zero ($V_B \approx 0V$), making the analysis straightforward.

and the voltages change to:
 $V_C = 10V - (2.15\text{ mA})(3.6\text{ k}\Omega) = 2.26V$
 and
 $V_{CE} = 2.26V - 1.1V = 1.16V$
 In this case, the Q point shifts to a new position at Q_H with coordinates of 2.15 mA and 1.16 V.

Two Supply Emitter Bias

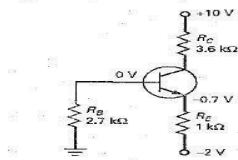
Two supply emitter bias of figure 8.8 shows a transistor circuit with two power supplies +10V and -2V. The negative supply forward biases the emitter diode. The positive supply reverse biases the collector diode. This circuit is derived from the emitter bias. Hence we refer this as two supply bias(TSEB).



The circuit in figure 8.8 is redrawn in figure 8.9 for simplicity. The steps for analysing circuit above as follows

1. If this circuit is correctly designed the base current I_B is very small and hence considered as approximately zero. Voltage V_B is approximately zero $V_B = 0$
2. Calculate the V_E which is obtained by analysing the circuit at emitter $V_E = V_B - V_{BE}$
3. Find the emitter current I_E
 $I_E = (V_E - (-V_{EE})) / R_E$
4. Assume the collector current is equal to emitter current.
 $I_C \approx I_E$
5. Calculate collector voltage using following equation.
 $V_C = V_{CC} - I_C R_C$
6. Then calculate collector emitter voltage (V_{CE}) by subtracting emitter voltage from collector voltage.
 $V_{CE} = V_C - V_E$

Figure 8-10 Base voltage is ideally zero.



$V_{BE} = -0.7V - (-2V) = 1.3V$
 Once you have found the voltage across the emitter resistor, calculate the emitter current with Ohm's law:
 $I_E = 1.3V / 1\text{ k}\Omega = 1.3\text{ mA}$
 This current flows through the 3.6 kΩ and produces a voltage drop that we subtract from +10 V as follows:
 $V_C = 10V - (1.3\text{ mA})(3.6\text{ k}\Omega) = 5.32V$
 The collector-emitter voltage is the difference between the collector voltage and the emitter voltage:
 $V_{CE} = 5.32V - (-0.7V) = 6.02V$

Figure: Two Supply Emitter Bias (TSEB) circuit and redrawn simplified form

2.1 TSEB Analysis – Step-by-Step

7. **$V_B \approx 0$:** Since the circuit is correctly designed, I_B is very small, so V_B is approximately zero.

$$V_B = 0V \text{ (approximately)}$$

8. **Calculate V_E :** Subtract the base-emitter voltage drop:

$$V_E = V_B - V_{BE} = 0 - 0.7 = -0.7V$$

9. **Calculate I_E :** The emitter current flows through R_E from V_E to the -VEE supply:

$$I_E = (V_E - (-V_{EE})) / R_E = (-0.7V - (-2V)) / 1\text{ k}\Omega = 1.3V / 1\text{ k}\Omega = 1.3\text{ mA}$$

10. **$I_C \approx I_E$:** Collector current approximately equals emitter current.

$$I_C \approx I_E = 1.3\text{ mA}$$

11. **Calculate V_C :** Collector voltage:

$$V_C = V_{CC} - I_C \times R_C = 10V - (1.3mA)(3.6k\Omega) = 5.32 V$$

12. **Calculate VCE:** Collector-emitter voltage:

$$V_{CE} = V_C - V_E = 5.32V - (-0.7V) = 6.02 V$$

Key Features of TSEB:

- Excellent Q-point stability due to negative supply fixing the emitter voltage
- No need for R1-R2 voltage divider at the base
- Base is directly connected through R_B to ground ($\approx 0V$)
- Used in differential amplifiers and operational amplifiers where dual supply is available

 **NOTE:** In TSEB, the voltage across R_E = V_E - (-V_{EE}) = -0.7 - (-2) = 1.3V. The negative supply effectively raises the voltage available across R_E, which stabilizes the emitter current.

3. Other Types of Bias

Other types of Bias

Emitter feedback bias

In base bias circuit of figure 8.12a the Q point is varies all over the load line due to replacement of transistor and change in temperature. Hence to stabilize Q point the emitter feedback bias of circuit shown in figure 8.12b used. Noticed that the emitter resistor is added at the circuit to stabilize the Q point.

Here are the equations for analysing the Q point

$$I_E = \frac{V_{CC} - V_{BE}}{R_E + R_B/\beta_{dc}}$$

$$V_E = I_E R_E$$

$$V_B = V_E + 0.7 \text{ V}$$

$$V_C = V_{CC} - I_C R_C$$

$$V_{CE} = V_C - V_E$$

Looking at equations above it is clear that when I_C increases, V_E increases causing V_B to increase. Which gives more V_B hence less voltage across R_B reduces base current I_B which opposes original increase in I_C . In this way change in I_C due to variation in temperature will be opposes by emitter feedback bias circuit. It is called feedback because of change in emitter voltage is being feedback to the base circuit.

Figure 8-12 (a) Base bias; (b) emitter-feedback bias.

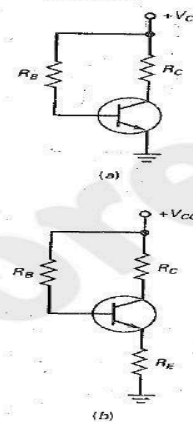
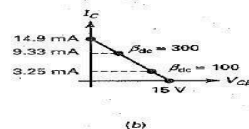
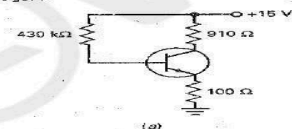


Figure 8-13 (a) Example of emitter-feedback bias; (b) Q point is sensitive to changes in current gain.



Collector – Feedback Bias

The figure 8.14a shows the collector feedback bias also called self-bias. Here feedback the voltage to the base from the collector to neutralize any change in collector current. For a instance suppose the collector current increases. This decreases the collector voltage V_C which decreases the voltage across the base resistor. In turn, this decreases the base current, which opposes the original increase in collector current I_C . Here are the equations for analysing collector feedback bias.

Figure: (a) Base bias circuit (b) Emitter-feedback bias circuit and load line

3.1 Emitter Feedback Bias

In a basic base bias circuit, the Q point wanders all over the load line due to transistor replacement and temperature changes (since it depends on beta). To stabilize the Q point, an emitter resistor R_E is added to the circuit — this is called Emitter Feedback Bias.

Feedback Mechanism:

When I_C increases (due to temperature rise or beta change):

- $V_E = I_E \times R_E$ increases (more current through R_E)
- $V_B = V_E + 0.7\text{V}$ also increases slightly
- This means MORE voltage is dropped across R_B from V_{CC} , so V_{BE} is slightly reduced
- Reduced $V_{BE} \rightarrow$ reduced $I_B \rightarrow$ reduced I_C (opposes the original increase)

This negative feedback action stabilizes the Q point. It is called "emitter feedback" because the change in emitter voltage is fed back to the base circuit to oppose the original change.

Analysis Equations:

$$I_E = (V_{CC} - V_{BE}) / (R_E + R_B/\beta_{dc})$$

$$V_E = I_E \times R_E$$

$$V_B = V_E + 0.7 \text{ V}$$

$$V_C = V_{CC} - I_C \times R_C$$

$$V_{CE} = V_C - V_E$$

✶ **NOTE:** Emitter feedback bias is better than base bias but not as stable as VDB. The Q point still depends on beta (β_{dc}), so it can shift with transistor replacement.

3.2 Collector Feedback Bias (Self-Bias)

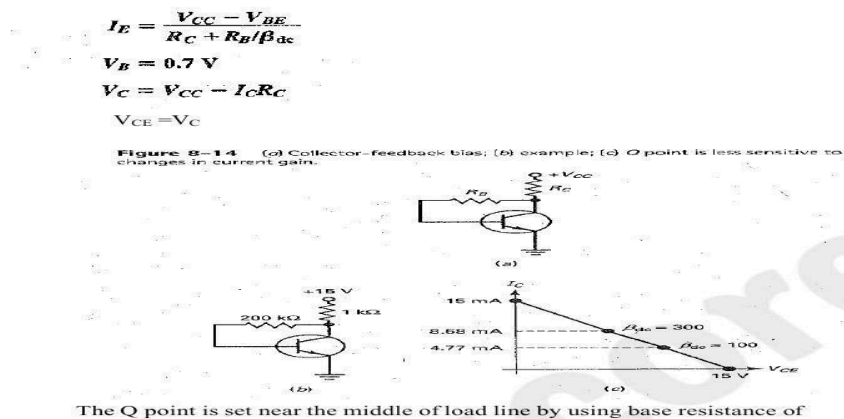
In Collector Feedback Bias (also called Self-Bias), the base resistor R_B is connected from the collector to the base (not from V_{CC} to base). This provides negative feedback directly from the collector voltage to the base.

Feedback Mechanism:

Suppose collector current I_C increases:

- V_C decreases (since $V_C = V_{CC} - I_C \times R_C$)
- Since R_B connects collector to base, lower V_C means lower V_B
- Lower $V_B \rightarrow$ lower $V_{BE} \rightarrow$ lower $I_B \rightarrow$ lower I_C (opposes the original increase)

This is self-stabilizing — the collector voltage automatically adjusts the base current to maintain stability.



$$R_B = \beta_{dc} R_C$$

Collector and Emitter Feedback bias

Collector and emitter feedback bias used to set more stable bias for transistor circuits. The figure 8.15 shows the collector – emitter feedback bias. Combining both types of feedback in one circuit helps to stabilize Q point but still production. Here are the equations for analysing it

$$I_E = \frac{V_{CC} - V_{BE}}{R_C + R_B + R_E/\beta_{dc}}$$

$$V_E = I_E R_E$$

$$V_B = V_E + 0.7 \text{ V}$$

$$V_C = V_{CC} - I_C R_C$$

Figure 8-15 Collector-emitter feedback bias.

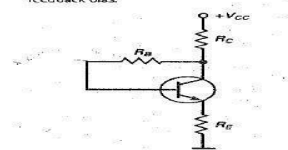


Figure: Collector feedback bias and Collector-Emitter feedback bias circuits

Analysis Equations:

$$I_E = (V_{CC} - V_{BE}) / (R_C + R_B/\beta_{dc})$$

$$V_B = 0.7 \text{ V}$$

$$V_C = V_{CC} - I_C \times R_C$$

$$V_{CE} = V_C \text{ (since emitter is at ground)}$$

Setting Q point near midpoint: The base resistance for midpoint bias:

$$R_B = \beta_{dc} \times R_C$$

3.3 Collector-Emitter Feedback Bias

This circuit combines BOTH collector feedback and emitter feedback in a single circuit to achieve even greater Q-point stability. The figure 8.15 shows this combined feedback configuration.

Analysis Equations:

$$I_E = (V_{CC} - V_{BE}) / (R_C + R_E + R_B / \beta_{dc})$$

$$V_E = I_E \times R_E$$

$$V_B = V_E + 0.7 \text{ V}$$

$$V_C = V_{CC} - I_C \times R_C$$

Bias Type	Q-point Stability	Depends on Beta?	Use Case
Base Bias	Poor — wanders with beta	Yes (strongly)	Simple circuits, fixed beta
Emitter Feedback	Moderate	Yes (less)	Improved base bias
Collector Feedback	Good	Somewhat	Self-biased circuits
VDB	Excellent	No (negligible)	Most practical amplifiers
TSEB	Excellent	No	Dual-supply amplifiers
Collector-Emitter Feedback	Very Good	Somewhat	High stability single-supply

PART B: BJT AC MODELS

4. Base Biased Amplifier

4.1 DC Biasing Circuit Analysis

A base biased amplifier uses a single resistor R_B between V_{CC} and the base to set the operating point. While simple, the Q point depends heavily on beta.

Base Biased Amplifier

DC biasing circuit

Figure 9.4 shows a base biased circuit, looking the circuit.

$$I_B = (30\text{V} - 0.7\text{V}) / 1\text{M} = 30\mu\text{A}$$

With current gain $\beta = 100$

$$I_C = I_B \times 100 = 3\text{mA}$$

And the collector voltage

$$V_C = V_{CC} - I_C \times R_C = 30 - (3\text{mA})(5\text{k}\Omega) = 15\text{V}$$

Hence the Q point is located at 3mA and 15V of I_C and V_C respectively

Amplifying circuit

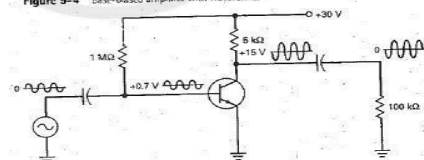
Figure 9.4 shows base bias amplifier circuit with an ac source voltage is 100 μV . Since the input coupling capacitor is an ac short, all the ac of source voltage appears between the base and the ground. For coupling capacitor work properly, its reactance must be 10 times less than input resistance of amplifier at the lowest frequency of the ac source ($X_C < 0.1R$). The ac voltage produces an ac base current that is added to the existing dc base current. Figure 9.3 a illustrates the ac component is super imposed on the dc component of 30 μA . The ac base current produces an amplified variation in collector current because of the current gain. In figure 9.3 b shows the amplified the ac collector current superimposed on dc collector of 3mA

Since this amplified collector current flows through the collector resistor, it produces a varying voltage across the collector resistor. Which is subtracted from the supply voltage we get the collector voltage shown in figure 9.3c.

Voltage waveforms

The voltage gain of an amplifier is defined as the ac output voltage divided by the ac input voltage.

Figure 9-4 Base-biased amplifier with waveforms.



where it is superimposed on the dc component of +0.7V. The variation in base voltage produces sinusoidal variation in base current, collector current and collector voltage. The total collector voltage is an inverted sine wave superimposed on the dc collector voltage of

Figure 9-3 DC and ac components. (a) Base current; (b) collector current; (c) collector voltage.

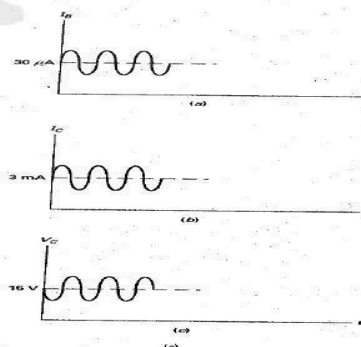


Figure 9.4 shows the waveforms for a base biased amplifier. The ac source voltage is small sinusoidal voltage. This is coupled into the base,

Figure: Base biased amplifier with DC/AC waveforms showing current and voltage components

DC Analysis Example ($V_{CC} = 30\text{V}$, $R_B = 1\text{M}\Omega$, $R_C = 5\text{k}\Omega$, $\beta = 100$):

$$I_B = (V_{CC} - V_{BE}) / R_B = (30\text{V} - 0.7\text{V}) / 1\text{M}\Omega = 30\mu\text{A}$$

$$I_C = \beta \times I_B = 100 \times 30\mu\text{A} = 3\text{mA}$$

$$V_C = V_{CC} - I_C \times R_C = 30\text{V} - (3\text{mA})(5\text{k}\Omega) = 15\text{V}$$

The Q point is at $I_C = 3\text{mA}$ and $V_C = 15\text{V}$ (midpoint of load line — ideal for amplification).

4.2 Amplifying Circuit & Voltage Waveforms

When an AC source voltage is applied (e.g., 100 μV) through an input coupling capacitor, the AC signal rides on top of the DC bias:

- The coupling capacitor acts as an AC short circuit — all AC source voltage appears between base and ground
- For coupling capacitor to work properly: $X_C < 0.1R$ (reactance at least 10x less than input resistance)
- The AC voltage produces an AC base current that adds to the existing DC base current (30 μ A DC)
- This AC base current is amplified by beta, producing an amplified AC collector current superimposed on the 3mA DC
- The amplified collector current flows through R_C , creating an amplified AC voltage at the collector
- The coupling capacitor at the output blocks the DC component — only AC reaches the load

Key Observation — Phase Inversion: The collector voltage is an INVERTED sine wave (180° phase shift) because when I_B increases $\rightarrow I_C$ increases $\rightarrow I_C \times R_C$ increases $\rightarrow V_C$ decreases. This inversion is the hallmark of a CE amplifier.

5. Emitter Biased Amplifier (VDB Amplifier)

5.1 VDB Amplifier with Bypass Capacitor

A VDB amplifier is the practical implementation of a common-emitter amplifier using voltage divider bias. To achieve high voltage gain, a bypass capacitor is added between the emitter and ground.

+15V. The coupling capacitor at the load blocks all dc component and hence the load voltage is pure ac signal.

Emitter Biased Amplifier

VDB(Voltage divider bias) Amplifier

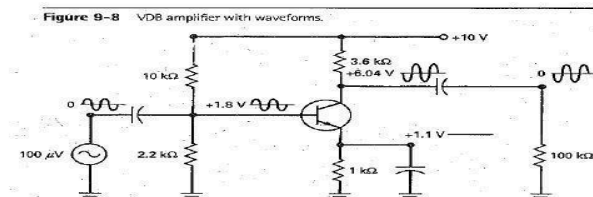


Figure 9.8 shows a voltage divider biased (VDB) amplifier. To calculate the dc voltages and currents, mentally open all capacitors. The quiescent or dc values for this circuit are

$$V_B = 10 \times 2.2k / 12.2k = 1.8V,$$

$$V_E = V_B - V_{BE} = 1.8 - 0.7 = 1.1V,$$

$$I_C = I_E = V_E / R_E = 1.1 / 1k = 1.1mA,$$

$$V_C = V_{CC} - I_C \times R_C = 10 - 1.1m \times 3.6k = 6.04V,$$

With coupling capacitors it is also use a bypass capacitor between the emitter and ground without this capacitor the ac base current would be much smaller. With this capacitor we get a much larger voltage gain. To work bypass capacitor properly its reactance must be minimum 10 times less than the resistance at lowest frequency of ac signal ($X_c < 0.1R$).

VDB Waveforms

Notice the voltage waveforms in figure 9.8. The ac source voltage is a small sinusoidal voltage with an average value of zero. The base voltage is an ac voltage superimposed on a dc voltage 1.8 V. The collector voltage is an amplified and inverted ac voltage superimposed on the dc collector voltage of +6.04V. The load voltage is the same as the collector voltage except that it has an average value of zero. It is also note that voltage at emitter is pure dc of +1.1V because of bypass capacitor.

Two- supply emitter bias (TSEB) amplifier

Figure 9.9 shows TSEB amplifier, The dc analysis of this circuit shown in figure 8.10.

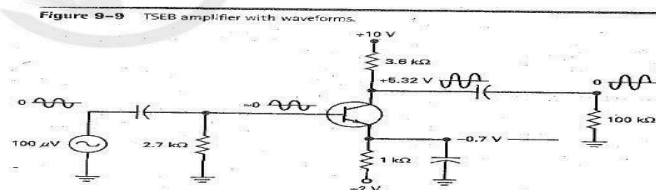


Figure shows a two supply emitter bias (TSEB) amplifier. By analysing the dc part of the circuit and calculate these quiescent voltages

Figure: VDB amplifier and TSEB amplifier with complete waveforms

DC Analysis ($V_{CC}=10V$, $R_1=10k\Omega$, $R_2=2.2k\Omega$, $R_C=3.6k\Omega$, $R_E=1k\Omega$):

$$V_B = 10V \times 2.2k / (10k + 2.2k) = 1.8V$$

$$V_E = V_B - 0.7V = 1.8 - 0.7 = 1.1V$$

$$I_C = I_E = V_E / R_E = 1.1V / 1k\Omega = 1.1mA$$

$$V_C = V_{CC} - I_C \times R_C = 10 - 1.1m \times 3.6k = 6.04V$$

Role of Bypass Capacitor:

- The bypass capacitor CE connects from emitter to ground
- For DC: CE is open — RE appears in circuit, stabilizing the bias point
- For AC: CE is short — RE is bypassed, so AC emitter is effectively grounded
- With CE: voltage gain is much larger (AC current does not see RE)
- Without CE: voltage gain is much smaller (AC current flows through RE, causing negative feedback)

- Bypass capacitor requirement: $X_C < 0.1 \times R_E$ at the lowest frequency of operation

VDB Waveforms (Figure 9.8):

- AC source: small sinusoidal voltage with average = 0
- Base voltage: AC riding on DC level of +1.8V
- Collector voltage: amplified, INVERTED AC riding on DC +6.04V
- Load voltage: same amplified AC with no DC (coupling cap removes DC)
- Emitter voltage: pure DC = +1.1V (because bypass cap shorts AC at emitter)

5.2 Two Supply Emitter Bias Amplifier

Figure 9.9 shows a TSEB amplifier using +10V and -2V supplies. The DC analysis gives:

$$V_B = 0V, V_E = -0.7V, I_C = 1.3mA, V_C = 5.32V$$

The TSEB amplifier has two coupling capacitors (at input and output) and a bypass capacitor. Waveforms: base voltage has tiny AC on $\approx 0V$ DC; total collector voltage is inverted sine on +5.32V DC; load voltage is amplified AC with no DC.

$$V_B = 0V, V_E = V_B - V_{BE} = -0.7V,$$

$$I_C = I_E = V_E - (-2)/R_E = -0.7 - (-2)/1k = 1.3mA,$$

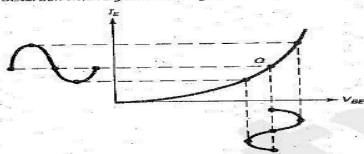
$$V_C = V_{CC} - I_C \times R_C = 10 - 1.3m \times 306k = 5.32V$$

Figure 9.9 shows two coupling capacitors to block ac at input and output side and a bypass capacitor to bypass ac component through emitter resistor R_E so that it does not disturb the operating point.

Notice waveform that the ac source voltage is a small sinusoidal voltage. The base voltage has a small ac component riding on a dc component of approximately 0v. The total collector voltage is inverted sine wave riding on the dc the dc collector voltage of +5.32V. The load voltage is the same amplified signal with no dc component.

1. Small Signal Operation

Figure 9-10 Distortion when signal is too large.



When an ac voltage is coupled into the base of a transistor, an ac voltage appears across the base emitter diode. This produces the sinusoidal variation in V_{BE} shown in figure 9.10. When the voltage increases to its positive peak, the instantaneous operating point moves from Q to the upper point. On the other hand when the sine wave decreases to its negative peak, the operating point moves from Q to the lower point.

Distortion & Reducing distortion

The ac emitter current is not a perfect replica of the ac base voltage. Since the graph is curved upward, the positive half cycle of ac emitter current is stretched and the negative half cycle is compressed. This stretching and compressing is called distortion.

One way to reduce this distortion is by keeping the ac base voltage small. When you reduce the peak value of base voltage, you reduce the movement of the instantaneous operating point. Using 10 percent rule to minimize the distortion, the peak to peak ac value of i_e must be small compared to I_{EQ} i.e. $(i_{e(pp)}) < 0.1I_{EQ}$. Where i_e is ac emitter current and I_{EQ} is dc emitter current.

AC Beta

The dc current gain (dc beta) is

$$\beta_{dc} = I_C / I_B$$

Where ac current gain is (ac beta) equal the ac collector current divided by the ac base current.

$$\beta = i_c / i_b$$

Graphically, β equals the slope the curve at point Q

Figure 9-12 AC current gain equals ratio of changes.

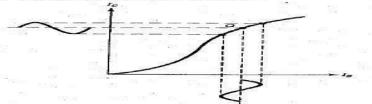


Figure: TSEB amplifier waveforms and Small Signal operation showing distortion

6. Small Signal Operation

When an AC voltage is coupled into the base of a transistor, an AC voltage appears across the base-emitter junction (VBE). This AC signal causes the instantaneous operating point to move up and down the IE vs VBE characteristic curve around the Q point.

Operating Point Movement:

- When AC voltage increases to positive peak: operating point moves FROM Q to the upper point (IC increases)
- When AC voltage decreases to negative peak: operating point moves FROM Q to the lower point (IC decreases)
- The result: IC and IE vary sinusoidally around their DC quiescent values

6.1 Distortion & Reducing Distortion

The IE-VBE characteristic curve is NOT linear — it is exponential in nature (like a diode curve). Because of this curvature, the AC emitter current is NOT a perfect replica of the AC base voltage:

- The positive half cycle of AC emitter current is STRETCHED (larger amplitude)
- The negative half cycle of AC emitter current is COMPRESSED (smaller amplitude)
- This asymmetric stretching and compressing is called DISTORTION (specifically, harmonic distortion)

How to Reduce Distortion — The 10% Rule:

One way to reduce distortion is by keeping the AC base voltage SMALL. When you reduce the peak value of the AC base voltage, you reduce the movement of the instantaneous operating point around Q, keeping it on the more linear portion of the curve.

Condition for small-signal operation: $i_{e(pp)} < 0.1 \times I_{EQ}$

Where: $i_{e(pp)}$ = peak-to-peak AC emitter current, I_{EQ} = DC quiescent emitter current. When this condition is satisfied, the amplifier operates in the LINEAR region and distortion is minimised.

 **NOTE:** The 10% rule is the fundamental condition for "small-signal" operation. As long as the AC signal swing is less than 10% of the DC operating point, the transistor behaves approximately linearly.

7. AC Beta

There are two distinct current gain parameters for a BJT transistor — DC beta and AC beta. Understanding the difference is essential for proper amplifier analysis.

DC Beta (β_{dc}):

DC beta is the ratio of DC collector current to DC base current at a given operating point:

$$\beta_{dc} = I_C / I_B$$

AC Beta (β or β_{ac}):

AC beta is the ratio of the AC (incremental) collector current to the AC (incremental) base current at the Q point:

$$\beta = i_c / i_b \quad (\text{where } i_c \text{ and } i_b \text{ are AC small-signal currents})$$

Graphically, AC beta equals the SLOPE of the I_C vs I_B curve at the Q point. Because the curve is slightly nonlinear, β_{ac} and β_{dc} are not exactly equal — β_{ac} can differ from β_{dc} , especially at low and high current levels.

Parameter	Formula	Meaning	Value Range
β_{dc} (DC beta / h_{FE})	I_C / I_B	DC current gain at Q point	20 to 500+ (depends on transistor and I_C)
β (AC beta / h_{fe})	i_c / i_b	AC small-signal current gain	Usually close to β_{dc} for small signals
Relation	$\beta \approx \beta_{dc}$	For small signals and typical operating points	Used interchangeably in most analyses

8. AC Resistance of the Emitter Diode (re')

When a small AC voltage v_{be} appears across the base-emitter junction, it produces an AC emitter current i_e . The ratio of this AC voltage to the AC current is defined as the AC emitter resistance r_e' (pronounced 'r-e-prime').

AC Resistance of the Emitter Diode

When a small ac voltage v_{be} change at the emitter and base junction produces the ac emitter current as shown in figure 9.13

The ac emitter resistance of the emitter diode is the ac base emitter voltage divided by the ac emitter current.

$$r_e' = v_{be} / i_e$$

The ac emitter resistance is also 25mV divided by dc emitter current I_E

$$r_e' = 25 \text{ mV} / I_E$$

Two Transistor Models

To analyse the ac operation of a transistor amplifier, we need an ac equivalent circuit(ac model) for a transistor that simulates how it behaves when an ac signal is present.

The T Model

The T model also called Ebers-Moll model shown in figure 9.16. Here the emitter diode of a transistor acts like an ac resistance r_e' and the collector diode acts like a current source i_c . Since it looks like T shape so it is called T model.

When an ac input signal is applied this gives an ac base emitter v_{be} across emitter base and produces ac base current i_b as shown in figure 9.17a. Hence Looking at base the input impedance $Z_{in}(\text{base})$ as shown in figure 9.17b is

$$Z_{in}(\text{base}) = \frac{v_{be}}{i_b} \quad (9-11)$$

Figure 9-16 T model of a transistor.



Figure 9-17 Defining the input impedance of the base.



Figure: AC resistance of emitter diode showing the relationship between v_{be} and i_e

Definition:

$$r_e' = v_{be} / i_e$$

Practical Formula:

The AC emitter resistance can be calculated from the DC emitter current I_E using the following important formula:

$$r_e' = 25 \text{ mV} / I_E$$

This formula is derived from the diode equation. At room temperature (25°C), the thermal voltage $V_T = 25\text{mV}$. The 25mV in the numerator is the thermal voltage of the emitter diode at room temperature.

Example Calculations:

DC Emitter Current I_E	AC Resistance r_e'	Significance
1 mA	$r_e' = 25\text{mV}/1\text{mA} = 25 \Omega$	Higher $r_e' \rightarrow$ lower voltage gain
5 mA	$r_e' = 25\text{mV}/5\text{mA} = 5 \Omega$	Lower $r_e' \rightarrow$ higher voltage gain
0.5 mA	$r_e' = 25\text{mV}/0.5\text{mA} = 50 \Omega$	Very low $I_E \rightarrow$ very low voltage gain

DC Emitter Current I_E	AC Resistance r_e'	Significance
10 mA	$r_e' = 25\text{mV}/10\text{mA} = 2.5 \Omega$	High $I_E \rightarrow$ maximum voltage gain

 **NOTE:** r_e' is the cornerstone of BJT small-signal analysis. It connects the DC bias point (I_E) to the AC gain ($A_v = r_c/r_e'$). To get high voltage gain, set I_E high (which reduces r_e').

9. Two Transistor Models

To analyse the AC operation of a transistor amplifier, we need an AC equivalent circuit (AC model) that accurately represents the transistor's behavior for small AC signals. Two models are commonly used: the T model and the pi (π) model.

Why AC Models Are Needed:

- DC supply sources become AC short circuits (replaced by wires in AC analysis)
- Capacitors become AC short circuits (they block DC but pass AC freely)
- The transistor itself must be replaced by its equivalent AC circuit
- The resulting simplified circuit can then be analysed using basic circuit laws (Ohm's law, KVL, KCL)

9.1 The T Model (Ebers-Moll Model)

The T model (also called the Ebers-Moll model) represents the transistor's AC behavior using the following components:

- Emitter diode is replaced by its AC resistance r_e'
- Collector diode is replaced by a current-controlled current source $i_c = \beta \times i_b$ (or $i_c \approx i_e$ for large β)
- The shape of the circuit resembles the letter T — hence the name T model

AC Resistance of the Emitter Diode

When a small ac voltage v_{be} change at the emitter and base junction produces the ac emitter current as shown in figure 9.13

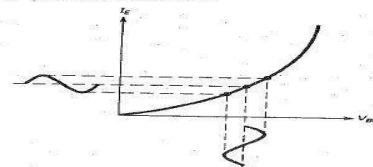
The ac emitter resistance of the emitter diode is the ac base emitter voltage divided by the ac emitter current.

$$r_e' = v_{be} / i_e$$

The ac emitter resistance is also 25mV divided by dc emitter current I_E

$$r_e' = 25 \text{ mV} / I_E$$

Figure 9-13 AC resistance of emitter diode.



Two Transistor Models

To analyse the ac operation of a transistor amplifier, we need an ac equivalent circuit (ac model) for a transistor that simulates how it behaves when an ac signal is present.

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$$Z_{in}(\text{base}) = \frac{v_{be}}{i_b} \quad (9-11)$$

Figure 9-16 T model of a transistor.



Figure 9-17 Defining the input impedance of the base.



Figure: T model and pi model of a BJT transistor showing input impedance at the base

Input Impedance of the Base (T Model):

When an AC input signal v_{be} is applied, it produces AC base current i_b . The input impedance looking into the base is:

$$Z_{in}(\text{base}) = v_{be} / i_b$$

Since $v_{be} = i_e \times r_e'$ and $i_e \approx \beta \times i_b$ (because $i_e = i_c + i_b \approx i_c = \beta i_b$):

$$Z_{in}(\text{base}) = (i_e \times r_e') / i_b = (\beta \times i_b \times r_e') / i_b = \beta \times r_e'$$

 **NOTE:** This is a key result: the input impedance looking into the base is β times r_e' . A transistor with $\beta=100$ and $r_e'=25\Omega$ has $Z_{in}(\text{base}) = 100 \times 25 = 2500\Omega = 2.5k\Omega$.

9.2 The pi (π) Model

The pi model is another equivalent circuit for the transistor, named because the circuit resembles the Greek letter π . It is generally EASIER to use than the T model because the input impedance ($\beta r_e'$) appears explicitly as a single element.

T Model to π Model Conversion:

Applying Ohm's law to the emitter diode in the T model:

$$v_{be} = i_e \times r_e'$$

Substituting into the input impedance equation and using $i_e \approx i_c = \beta \times i_b$:

$$Z_{in}(\text{base}) = v_{be}/i_b = (i_e \times r_e')/i_b = (\beta \times i_b \times r_e')/i_b = \beta \times r_e'$$

Therefore, the π model replaces the base-emitter junction with a single resistance $\beta \times r_e'$ between base and emitter, and retains the current source $i_c = \beta \times i_b$ at the collector.

Feature	T Model	Pi Model
Circuit shape	T-shaped (r_e in emitter branch)	Pi-shaped (βr_e in base branch)
Input impedance	Not immediately obvious from circuit	Immediately visible as βr_e
Ease of use	Less intuitive for Z calculations	More intuitive - βr_e directly visible
Current source	$i_c = \beta i_b$ at collector	$i_c = \beta i_b$ at collector (same)
When preferred	Voltage gain derivation	Input/output impedance analysis

10. Analyzing an Amplifier (DC + AC Analysis)

The standard approach to fully analyse a transistor amplifier is to split the analysis into two independent parts:

Step 1 – DC Analysis: OPEN all capacitors. Analyse the resulting DC equivalent circuit to find the Q point (I_C , V_{CE} , I_E , V_B , V_E , V_C).

Step 2 – AC Analysis: SHORT all capacitors and DC voltage sources (replace with wires). Replace the transistor with its π model or T model. Analyse the resulting AC equivalent circuit for voltage gain, input impedance, and output impedance.

10.1 Base Bias Amplifier Analysis

DC Analysis: Open all capacitors. Calculate I_B , I_C , V_C for the base bias configuration.

AC Equivalent Circuit (Figure 9.20b): Short all capacitors and VCC. Replace transistor with π model.

- In the base circuit: AC input voltage v_{in} appears across R_B in parallel with $\beta r_e'$
- In the collector circuit: current source pumps AC current i_c through R_C in parallel with R_L
- Input impedance: $Z_{in}(\text{base}) = \beta r_e'$; $Z_{in}(\text{stage}) = R_B \parallel \beta r_e'$

The π Model

Figure 9.18a shows the π model of transistor. This model is easier to use than the T model shown in figure 9.18b because the input impedance is not obvious in T model.

Applying ohm's law to the emitter diode of fig. 9.17a, we can write,

$$v_{be} = i_e r_e'$$

Equating $v_{be} = i_e r_e'$ into equation 9.11 will give

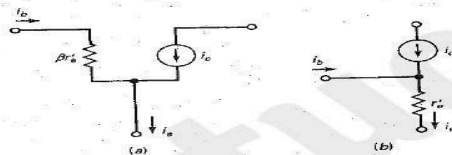
$$Z_{in}(\text{base}) = \frac{v_{be}}{i_b} = \frac{i_e r_e'}{i_b}$$

Since $i_e \approx i_c$, the foregoing equation simplifies to:

$$Z_{in}(\text{base}) = \beta r_e'$$

Hence π model shown in figure 9.18a having input impedance of $\beta r_e'$

Figure 9-18 π model of a transistor.



Analysing an Amplifier

The simplest way to analyse an amplifier is to split the analysis into two parts, 1) dc analysis
b) ac analysis

In dc analysis we calculate the dc voltages and currents to do this we need to open all capacitors. The circuit that remains is dc equivalent circuit.

In ac analysis short all capacitors and dc voltage sources and then transistor can be replaced by either the π model or the T model.

Analysis of Base bias amplifier

Figure 9.20a shows base biased amplifier. For dc analysis open all capacitors and analyse the dc equivalent circuit which means calculate the operating point values I_C and V_{CE} .

For ac analysis the figure 9.20b shows the ac equivalent circuit. Which we got after shorting all capacitors and dc source +VCC and then replace the transistor with its π model. As you can see in the base circuit, the ac input voltage appears across R_B in parallel with $\beta r_e'$. In the collector circuit, the current source pumps an ac current of i_c through R_C in parallel with R_L .

Figure: Pi model, analysis of base bias and VDB amplifier AC equivalent circuits

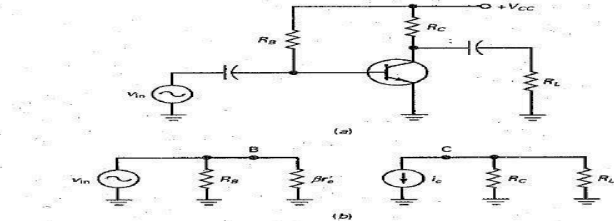
10.2 VDB Amplifier Analysis

DC Analysis: Open all capacitors. Calculate V_B , V_E , I_C , V_C using voltage divider method.

AC Equivalent Circuit (Figure 9.21b): Short all capacitors and $V_{CC} \rightarrow R_1$ appears from base to AC ground; V_{CC} rail becomes AC ground.

- In the base circuit: v_{in} appears across $R_1 \parallel R_2 \parallel \beta r_e'$
- In the collector circuit: current source i_c flows through $R_C \parallel R_L$
- Input impedance: $Z_{in}(\text{stage}) = R_1 \parallel R_2 \parallel \beta r_e'$

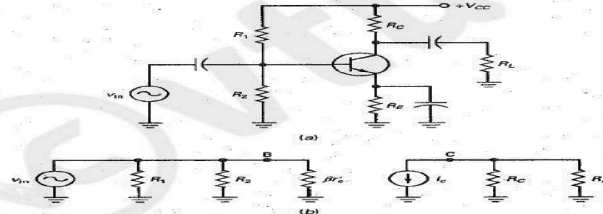
Figure 9-20 (a) Base-biased amplifier; (b) ac equivalent circuit.



Analysis of Voltage divider bias amplifier

Figure 9.21a is VDB amplifier and figure 9.21b is its ac equivalent circuit. As you can see. All capacitors have been shorted, the dc supply point has become an ac ground and the transistor has been replaced with $\beta r_e'$ in parallel with βr_e . In the collector circuit, the current source pumps an ac current of i_c through R_C in parallel with R_L .

Figure 9-21 (a) VDB amplifier; (b) ac equivalent circuit.



Analysis of Two supply emitter bias amplifier

The two supply emitter biasing is shown in figure 9.22a. The equivalent for TSEB is shown in figure 9.22b to get it again all capacitors and dc voltage source are shorted and the transistor is replaced by its π model. In the base circuit, the ac input voltage appears across R_B in parallel with $\beta r_e'$. In the collector circuit, the current source pumps an ac current of i_c through R_C in parallel with R_L .

Figure: VDB amplifier and TSEB amplifier AC equivalent circuits

10.3 TSEB Amplifier Analysis

DC Analysis: Open capacitors. $V_B = 0$, $V_E = -0.7V$, $I_C = 1.3mA$, $V_C = 5.32V$.

AC Equivalent Circuit (Figure 9.22b): Short all capacitors and both supply rails.

- In the base circuit: v_{in} appears across $R_B \parallel \beta r_e'$
- In the collector circuit: current source i_c flows through $R_C \parallel R_L$
- Input impedance: $Z_{in}(\text{stage}) = R_B \parallel \beta r_e'$

11. H Parameters & Relation with R Parameters

When the transistor was first invented, H parameters (Hybrid parameters) were the standard method to analyse transistor circuits. Today, the r-parameter method (using β and r_e') is more practical and widely used. However, H parameters still appear on datasheets and must be understood.

The Four H Parameters:

H Parameter	Symbol	Physical Meaning	Relation to R Parameters
Forward current gain	h_{fe} (or h_{FE})	AC current gain ($=\beta$) in CE configuration	$\beta = h_{fe}$
Input impedance	h_{ie}	Input impedance at base	$h_{ie} = \beta \times r_e' = h_{fe} \times r_e'$
Reverse voltage ratio	h_{re}	Feedback voltage ratio (internal)	Usually negligible ($\sim 10^{-4}$)
Output admittance	h_{oe}	Output conductance at collector	Usually negligible for basic design

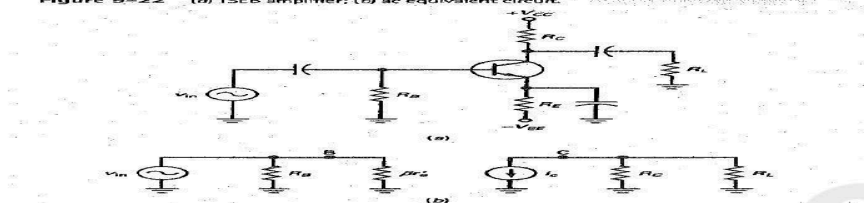
Key Relations:

$$\beta = h_{fe} \quad (\text{AC current gain})$$

$$r_e' = h_{ie} / h_{fe} \quad (\text{AC emitter resistance})$$

Practical Note: Only h_{fe} and h_{ie} are needed for basic amplifier design and troubleshooting. The parameters h_{re} and h_{oe} represent internal feedback and output conductance effects that are negligible in most low-frequency designs.

Figure 9-22 (a) TSEB amplifier; (b) ac equivalent circuit.



H parameters

When transistor was first invented, an approach used to analyse and design the transistor circuit is known as h parameters. The more practical approach is the one using π model which is called r-parameter method, which uses the quantity like β and r_e' but h parameters are still survived on data sheet. There are h parameters which gives very useful information are h_{ie} , h_{re} and h_{oe} .

Relation between R and H parameters

1. The h_{fe} in H parameter is nothing but a current gain β in R parameters
 $\beta = h_{fe}$
2. Similarly the quantity h_{ie} in H parameter equivalent to the input impedance which is relate to R parameter's r_e' like this
 $r_e' = \frac{h_{ie}}{h_{fe}}$
3. The another two parameters h_{re} and h_{oe} are not needed for troubleshooting and basic design

Voltage Amplifiers

Voltage gain

Figure 10.1a shows a voltage divider biased amplifier. Here voltage gain was defined as the ac output voltage divided by the ac input voltage. Let us derive the voltage gain equation using both T and π model

Derived from the π Model

Figure 10-1b shows the ac equivalent circuit using the π model of the transistor. The ac base current i_b flows through the input impedance of the base ($\beta r_e'$). With Ohm's law, we can write:

$$V_{in} = i_b \beta r_e'$$

In the collector circuit, the current source pumps an ac current i_c through the parallel connection of R_C and R_L . Therefore, the ac output voltage equals:

$$V_{out} = i_c (R_C \parallel R_L) = \beta i_b (R_C \parallel R_L)$$

Figure: H parameter equivalent circuit and R-H parameter relations

PART C: VOLTAGE AMPLIFIERS**12. Voltage Gain Derivation**

Voltage gain (A_v) is defined as the ratio of AC output voltage to AC input voltage. It is the most important figure of merit for a voltage amplifier.

$$A_v = v_{out} / v_{in}$$

12.1 Voltage Gain Derived from π Model

Using the π model (Figure 10.1b), the AC base current i_b flows through the input impedance $\beta r_e'$ of the base. Applying Ohm's law to the input:

$$v_{in} = i_b \times \beta \times r_e'$$

In the collector circuit, the current source pumps AC current i_c through the parallel combination of R_C and R_L . The AC output voltage equals:

$$v_{out} = i_c \times (R_C \parallel R_L) = \beta \times i_b \times (R_C \parallel R_L)$$

Dividing v_{out} by v_{in} :

$$A_v = v_{out}/v_{in} = \beta \times i_b \times (R_C \parallel R_L) / (i_b \times \beta \times r_e') = (R_C \parallel R_L) / r_e'$$

Defining ac collector resistance r_c :

$$r_c = R_C \parallel R_L$$

Therefore, the voltage gain simplifies to:

$$A_v = r_c / r_e'$$

Now, we can divide v_{out} by v_{in} to get:

$$A_v = \frac{v_{out}}{v_{in}} = \frac{\beta i_b (R_C \parallel R_L)}{i_b \beta r_e'}$$

which simplifies to:

$$A_v = \frac{(R_C \parallel R_L)}{r_e'}$$

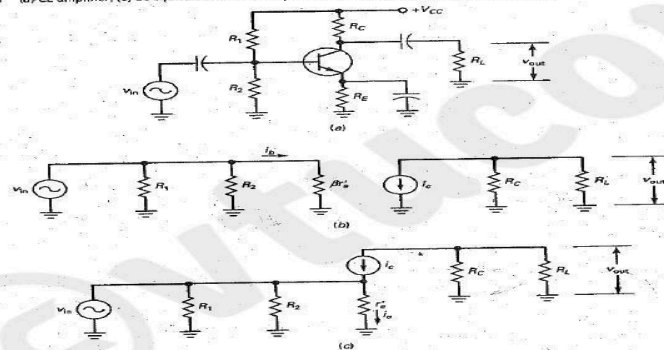
From the figure 10.1b The total load resistance seen by the collector is parallel combination of R_C and R_L . This is called the ac collector resistance r_c .

$$r_c = R_C \parallel R_L$$

Now, we can rewrite Eq. (10-1) as:

$$A_v = \frac{r_c}{r_e'}$$

Figure 10-1 (a) CE amplifier; (b) ac equivalent circuit with π model; (c) ac equivalent circuit with T model.



Derived from the T model

The figure 10.c shows the ac equivalent circuit using the T model of the transistor. The input voltage v_{in} appears across r_e' . With Ohm's law we can write:

Figure: CE amplifier (a) circuit (b) ac equivalent with π model (c) with T model

12.2 Voltage Gain Derived from T Model

Using the T model (Figure 10.1c), the input voltage v_{in} appears directly across r_e' . Applying Ohm's law:

$$v_{in} = i_e \times r_e'$$

In the collector circuit, the current source pumps i_c through the AC collector resistance r_c . Therefore:

$$v_{out} = i_c \times r_c$$

Dividing v_{out} by v_{in} and using $i_c \approx i_e$:

$$A_v = v_{out}/v_{in} = i_c \times r_c / (i_e \times r_e') = r_c/r_e' \quad (\text{since } i_c \approx i_e)$$

Both models give the same result:

$$A_v = r_c / r_e' \quad (\text{Fundamental voltage gain equation})$$

$$v_{in} = i_e r_e'$$

In the collector circuit, the current source pumps an ac current i_c through the ac collector resistance. Therefore, the ac output voltage equals:

$$v_{out} = i_c r_c$$

Now, we can divide v_{out} by v_{in} to get:

$$A_v = \frac{v_{out}}{v_{in}} = \frac{i_c r_c}{i_e r_e'}$$

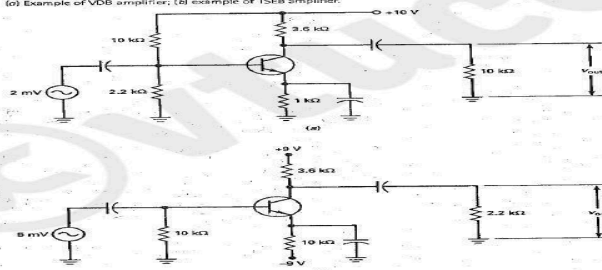
Since $i_c = i_e$, we can simplify the equation to get:

$$A_v = \frac{r_c}{r_e'}$$

Example 10-1

What is the voltage gain in Fig. 10-2a? The output voltage across the load resistor?

Figure 10-2 (a) Example of VDB amplifier; (b) example of TSEB amplifier.



SOLUTION The ac collector resistance is:

$$r_c = R_C \parallel R_L = (3.6 \text{ k}\Omega \parallel 10 \text{ k}\Omega) = 2.65 \text{ k}\Omega$$

In Example 9-2, we calculated an r_e' of 22.7Ω . So, the voltage gain is:

$$A_v = \frac{r_c}{r_e'} = \frac{2.65 \text{ k}\Omega}{22.7 \Omega} = 117$$

The output voltage is:

$$v_{out} = A_v v_{in} = (117)(2 \text{ mV}) = 234 \text{ mV}$$

Figure: Voltage gain calculation from T model derivation

Example 10-1: VDB amplifier, $R_C=3.6\text{k}\Omega$, $R_L=10\text{k}\Omega$, $r_e'=22.7\Omega$

$$r_c = R_C \parallel R_L = 3.6\text{k}\Omega \parallel 10\text{k}\Omega = 2.65 \text{ k}\Omega$$

$$A_v = r_c/r_e' = 2.65\text{k}\Omega / 22.7\Omega = 117$$

$$v_{out} = A_v \times v_{in} = 117 \times 2\text{mV} = 234 \text{ mV}$$

Example 10-2: TSEB amplifier, $R_C=3.6\text{k}\Omega$, $R_L=2.2\text{k}\Omega$, $V_{CC}=+9\text{V}$, $V_{EE}=-9\text{V}$, $R_B=10\text{k}\Omega$

$$r_c = 3.6\text{k}\Omega \parallel 2.2\text{k}\Omega = 1.37 \text{ k}\Omega$$

$$I_E = (9\text{V} - 0.7\text{V})/10\text{k}\Omega = 0.83 \text{ mA} \rightarrow r_e' = 25\text{mV}/0.83\text{mA} = 30 \Omega$$

$$A_v = 1.37\text{k}\Omega / 30\Omega = 45.7$$

$$v_{out} = 45.7 \times 5\text{mV} = 228 \text{ mV}$$

13. Loading Effect of Input Impedance

In a practical amplifier circuit, the AC source (signal generator) has an internal resistance R_G . This internal resistance forms a voltage divider with the amplifier's input impedance, causing some of the source voltage to be dropped across R_G before reaching the transistor base. This reduction of signal voltage is called the **LOADING EFFECT**.

Example 10-2
What is the voltage gain in Fig. 10-2b? The output voltage across the load resistor?

SOLUTION The ac collector resistance is:

$$r_c = R_C \parallel R_L = (3.6 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega) = 1.37 \text{ k}\Omega$$
 The dc emitter current is approximately:

$$I_E = \frac{9 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = 0.83 \text{ mA}$$
 The ac resistance of the emitter diode is:

$$r_e' = \frac{25 \text{ mV}}{0.83 \text{ mA}} = 30 \Omega$$
 The voltage gain is:

$$A_v = \frac{r_c}{r_e'} = \frac{1.37 \text{ k}\Omega}{30 \Omega} = 45.7$$
 The output voltage is:

$$v_{out} = A_v v_{in} = (45.7)(5 \text{ mV}) = 228 \text{ mV}$$

The loading effect of input impedance

Input impedance :

In figure 10.30a, an ac voltage source v_g has an internal resistance R_G . Some of the ac source voltage is dropped across its internal resistance. As a result, the ac voltage between the base and ground is less than ideal.

The ac generator has to drive the input impedance of the stage $Z_{in}(\text{stage})$. An equivalent circuit of figure 10.3a shown in figure 10.3b. Looking from the input circuit of fig 10.3b input impedance $Z_{in}(\text{stage})$ is given by.

$$Z_{in}(\text{stage}) = R_1 \parallel R_2 \parallel \beta r_e'$$

Therefore equation for input voltage V_{in} looking at figure 10.3c will be

$$v_{in} = \frac{Z_{in}(\text{stage})}{R_G + Z_{in}(\text{stage})} v_g$$

Figure: CE amplifier showing input impedance loading effect and voltage divider action

Stage Input Impedance:

Looking into the input of the amplifier stage (from the source), the input impedance of the stage is the parallel combination of all resistances at the base:

$$Z_{in}(\text{stage}) = R_1 \parallel R_2 \parallel \beta \times r_e'$$

Actual Input Voltage Due to Loading:

The actual voltage that appears at the transistor base (v_{in}) is reduced from the generator voltage (v_g) by the voltage divider formed by R_G and $Z_{in}(\text{stage})$:

$$v_{in} = Z_{in}(\text{stage}) / (R_G + Z_{in}(\text{stage})) \times v_g$$

Impact on Output Voltage:

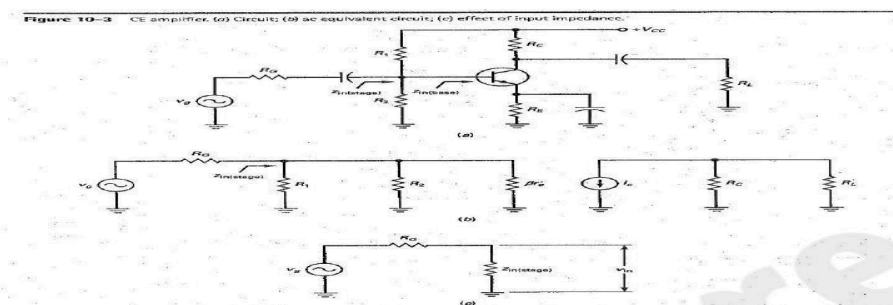
$$v_{out} = A_v \times v_{in} \text{ (not } A_v \times v_g \text{!)}$$

Example 10-3: $R_G=600\Omega$, $\beta=300$, $r_e'=22.7\Omega$, $A_v=117$, $v_g=2\text{mV}$, $R_1=10\text{k}\Omega$, $R_2=2.2\text{k}\Omega$

$$\begin{aligned}
 Z_{in(\text{base})} &= \beta \times r_e' = 300 \times 22.7 = 6.8 \text{ k}\Omega \\
 Z_{in(\text{stage})} &= 10 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \parallel 6.8 \text{ k}\Omega = 1.42 \text{ k}\Omega \\
 v_{in} &= 1.42 \text{ k}\Omega / (600 \Omega + 1.42 \text{ k}\Omega) \times 2 \text{ mV} = 1.41 \text{ mV} \\
 v_{out} &= 117 \times 1.41 \text{ mV} = 165 \text{ mV} \text{ (vs } 234 \text{ mV without loading)}
 \end{aligned}$$

Example 10-4: Same circuit but $\beta=50$:

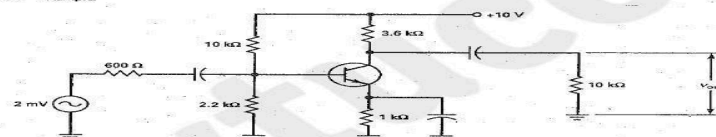
$$\begin{aligned}
 Z_{in(\text{base})} &= 50 \times 22.7 = 1.14 \text{ k}\Omega \\
 Z_{in(\text{stage})} &= 10 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \parallel 1.14 \text{ k}\Omega = 698 \Omega \\
 v_{in} &= 698 \Omega / (600 \Omega + 698 \Omega) \times 2 \text{ mV} = 1.08 \text{ mV}
 \end{aligned}$$



Example 10-3

In Fig. 10-4, the ac generator has an internal resistance of 600 Ω . What is the output voltage in Fig. 10-4 if $\beta = 300$?

Figure 10-4 Example.



SOLUTION Here are two quantities calculated in earlier examples: $r_e' = 22.7 \Omega$ and $A_V = 117$. We will use these values in solving the problem.

When $\beta = 300$, the input impedance of the base is:

$$Z_{in(\text{base})} = (300)(22.7 \Omega) = 6.8 \text{ k}\Omega$$

The input impedance of the stage is:

$$Z_{in(\text{stage})} = 10 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \parallel 6.8 \text{ k}\Omega = 1.42 \text{ k}\Omega$$

With Eq. (10-4), we can calculate the input voltage:

$$v_{in} = \frac{1.42 \text{ k}\Omega}{600 \Omega + 1.42 \text{ k}\Omega} 2 \text{ mV} = 1.41 \text{ mV}$$

This is the ac voltage that appears at the base of the transistor, equivalent to the ac voltage across the emitter diode. The amplified output voltage equals:

$$v_{out} = A_V v_{in} = (117)(1.41 \text{ mV}) = 165 \text{ mV}$$

Example 10-4

Repeat the preceding example for $\beta = 50$.

SOLUTION When $\beta = 50$, the input impedance of the base decreases to:

$$Z_{in(\text{base})} = (50)(22.7 \Omega) = 1.14 \text{ k}\Omega$$

The input impedance of the stage decreases to:

$$Z_{in(\text{stage})} = 10 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \parallel 1.14 \text{ k}\Omega = 698 \Omega$$

With Eq. (10-4), we can calculate the input voltage:

$$v_{in} = \frac{698 \Omega}{600 \Omega + 698 \Omega} 2 \text{ mV} = 1.08 \text{ mV}$$

Figure: Input loading effect examples – comparison for different beta values

NOTE: Lower beta $\rightarrow$ lower $Z_{in(\text{base})} \rightarrow$ lower $Z_{in(\text{stage})} \rightarrow$ more voltage divider loading $\rightarrow$ less $v_{in} \rightarrow$ less v_{out} . This demonstrates why high-beta transistors are preferred in voltage amplifiers: they reduce loading and preserve voltage gain.

PART D: CC AMPLIFIERS

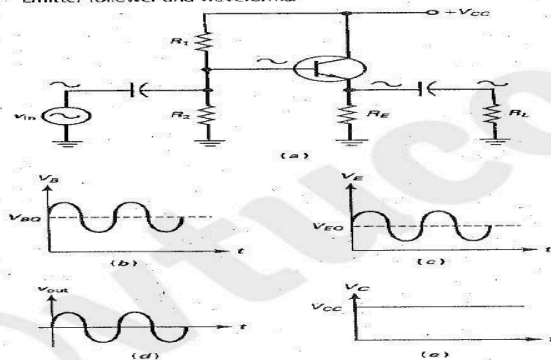
14. CC Amplifier (Emitter Follower)

The Common Collector (CC) amplifier is also called an EMITTER FOLLOWER because the output voltage at the emitter "follows" (tracks) the input voltage at the base. Unlike the CE amplifier, the output is taken from the EMITTER (not the collector), and the collector is AC grounded (connected to VCC which is AC ground).

CC (Common Collector Amplifier)

The emitter follower is also called a common collector (CC) amplifier. Figure 11.1a shows an emitter follower at which input voltage is coupled to the base and output produces at the emitter. The figure 11.1b shows the total voltage between base and ground. It has a dc component quiescent voltage V_{BQ} and an input voltage ac component rides on V_{BQ} . Similarly figure 11.1c shows the voltage between the emitter and ground. This time, the ac input voltage is entered on a quiescent emitter voltage V_{EQ} . The ac emitter output voltage coupled to the load resistor as shown in figure 11.1d. This output voltage is approximately equal to the input voltage. Because the output voltage follows the input voltage this circuit is called emitter follower.

Figure 11-1 Emitter follower and waveforms.

Voltage gain of CC amplifier:

As shown in figure 11.1a the total ac emitter resistance is

$$r_e = R_E \parallel R_L \quad (11-1)$$

Figure 11-2a shows the ac equivalent with the T model. Using Ohm's law, we can write these two equations:

$$v_{out} = i_e r_e$$

$$v_{in} = i_e (r_e + r'_e)$$

Divide the first equation by the second, and you get the voltage gain of the emitter follower:

$$A_v = \frac{r_e}{r_e + r'_e} \quad (11-2)$$

Figure: Emitter follower (CC amplifier) circuit and complete waveforms at all nodes

Key Characteristics of CC Amplifier:

Property	CE Amplifier	CC Amplifier (Emitter Follower)
Input terminal	Base	Base
Output terminal	Collector	Emitter
Common terminal	Emitter (bypassed)	Collector (AC ground)
Voltage gain	High (r_c/r'_e) $\gg 1$	Less than 1 (≈ 0.99)
Phase shift	180° (inverting)	0° (non-inverting)
Input impedance	Moderate ($\beta r'_e$)	High ($\beta(r_e + r'_e)$)
Output impedance	High ($\approx R_C$)	Very low

Property	CE Amplifier	CC Amplifier (Emitter Follower)
Primary use	Voltage amplification	Impedance matching / buffer

Why Called Emitter Follower: The output voltage at the emitter is approximately equal to the input voltage at the base. The emitter voltage tracks the base voltage with a voltage gain just below 1 (≈ 0.99). The output follows the input but with much lower impedance — this is perfect for driving low-impedance loads.

14.1 Voltage Gain of CC Amplifier

The total AC emitter resistance r_e (external) is the parallel combination of R_E and R_L :

$$r_e = R_E \parallel R_L$$

From the AC equivalent circuit using the T model (Figure 11-2a), applying Ohm's law:

$$v_{out} = i_e \times r_e \text{ and } v_{in} = i_e \times (r_e + r_e')$$

Dividing the first equation by the second gives the voltage gain of the emitter follower:

$$A_v = r_e / (r_e + r_e')$$

Since r_e is typically much larger than r_e' (e.g., $r_e = 1.66\text{k}\Omega$ vs $r_e' = 8\Omega$), the voltage gain is very close to 1 but always slightly less than 1.

Example 11-3 (Figure 11-5): $V_{CC}=+15\text{V}$, $R_1=R_2=4.7\text{k}\Omega$, $R_E=2.2\text{k}\Omega$, $R_L=6.8\text{k}\Omega$, $R_G=600\Omega$, $V_G=1\text{V}$, $\beta=150$

$$V_B = 15\text{V}/2 = 7.5\text{V} \rightarrow I_E = 6.8\text{V}/2.2\text{k}\Omega = 3.09\text{mA}$$

$$r_e' = 25\text{mV}/3.09\text{mA} = 8.09\Omega$$

$$r_e = 2.2\text{k}\Omega \parallel 6.8\text{k}\Omega = 1.66\text{k}\Omega$$

$$A_v = 1.66\text{k}\Omega / (1.66\text{k}\Omega + 8.09\Omega) = 0.995 \text{ } (\approx 1, \text{non-inverting})$$

14.2 Input Impedance of CC Amplifier

The input impedance looking into the base of the CC amplifier is:

$$Z_{in}(\text{base}) = \beta(r_e + r_e')$$

Note: this is β times the TOTAL emitter resistance ($r_e + r_e'$), NOT just $\beta \times r_e'$ as in the CE amplifier. This is why the CC amplifier has a MUCH higher input impedance than a CE amplifier.

The total input impedance of the stage (looking from the source) is:

$$Z_{in}(\text{stage}) = R_1 \parallel R_2 \parallel \beta(r_e + r_e')$$

From Example 11-3:

$$Z_{in}(\text{base}) = 150 \times (1.66\text{k}\Omega + 8.09\Omega) = 250\text{k}\Omega$$

$$Z_{in}(\text{stage}) = 4.7\text{k}\Omega \parallel 4.7\text{k}\Omega \parallel 250\text{k}\Omega \approx 2.35\text{k}\Omega$$

$$v_{in} = 2.35\text{k}\Omega / (600\Omega + 2.35\text{k}\Omega) \times 1\text{V} = 0.797\text{V}$$

$$v_{out} = 0.995 \times 0.797\text{V} = 0.793\text{V} \text{ (output very close to input!)}$$

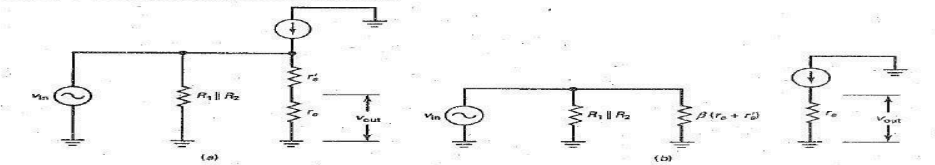
14.3 Output Impedance of CC Amplifier

Figure 11.8 shows the AC equivalent circuit for determining the output impedance of the emitter follower. The output impedance is found by looking back into the emitter terminal with R_L removed.

Considering the emitter current i_e flowing through the total resistance $R_G \parallel R_1 \parallel R_2$ at the input side, the output impedance Z_{out} is:

$$Z_{out} = R_E \parallel (r_e' + (R_G \parallel R_1 \parallel R_2) / \beta)$$

Figure 11-2 AC equivalent circuits for emitter follower.



Input impedance of the CC amplifier :

The figure 11.2b shows the ac equivalent circuit of CC amplifier looking in this figure input impedance of the base is

$$z_{in(base)} = \beta(r_e + r_e') \quad (11-3)$$

Hence the total input impedance of the input stage looking from the source Vin in figure 11.2b is

$$z_{in(stage)} = R_1 \parallel R_2 \parallel \beta(r_e + r_e') \quad (11-4)$$

Output impedance of the CC amplifier

Figure 11.8 shows ac equivalent circuit for emitter follower. Looking from the output load resistor RL and by considering the emitter current ie is flowing through the total resistor RG||R1||R2 at the input side, we have output impedance Zout is

$$z_{out} = R_E \parallel \left(r_e' + \frac{R_G \parallel R_1 \parallel R_2}{\beta} \right) \quad (11-5)$$

Figure 11-8 Output impedance of emitter follower.

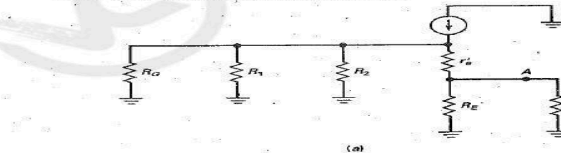


Figure: CC amplifier AC equivalent circuit for input/output impedance and Example 11-3

Key Insight: The output impedance is very LOW because the term $(R_G \parallel R_1 \parallel R_2)/\beta$ divides the source/bias resistance by beta. For large beta, this can be very small — giving an output impedance close to r_e' alone.

This is why the emitter follower is used as a BUFFER or IMPEDANCE CONVERTER:

- High input impedance: does not load the previous stage significantly
- Low output impedance: can drive low-impedance loads without significant signal loss
- Voltage gain ≈ 1 : passes signal faithfully
- No phase inversion: output is in phase with input

15. Quick Revision – All Topics at a Glance

 Complete exam-focused summary of all Module 1 topics.

15.1 Transistor Biasing – Summary

Bias Type	Key Formula	VB	Stability
Voltage Divider (VDB)	$V_{BB} = R_2 / (R_1 + R_2) \times V_{CC}$	Fixed by R_1, R_2	Excellent (no β dependence)
Two Supply Emitter (TSEB)	$I_E = (V_E - (-V_{EE})) / R_E$	$V_B \approx 0V$	Excellent
Emitter Feedback	$I_E = (V_{CC} - V_{BE}) / (R_E + R_B / \beta)$	Varies	Moderate
Collector Feedback	$I_E = (V_{CC} - V_{BE}) / (R_C + R_B / \beta)$	0.7V	Good
Collector-Emitter FB	$I_E = (V_{CC} - V_{BE}) / (R_C + R_E + R_B / \beta)$	$V_E + 0.7V$	Very Good

15.2 VDB Step-by-Step Formula Chain

$$V_{BB} = R_2 / (R_1 + R_2) \times V_{CC} \rightarrow V_E = V_{BB} - 0.7 \rightarrow I_E = V_E / R_E \rightarrow I_C \approx I_E \rightarrow V_C = V_{CC} - I_C R_C \rightarrow V_{CE} = V_C - V_E$$

15.3 AC Analysis – Key Formulas

Parameter	Formula	Notes
AC emitter resistance r_e'	$r_e' = 25mV / I_E$	Depends on DC bias point I_E
AC beta β	$\beta = i_c / i_b (\approx \beta_{dc} = h_{fe})$	Slope of I_C - I_B curve at Q
Input impedance (base) CE	$Z_{in}(base) = \beta \times r_e'$	For CE amplifier
Input impedance (base) CC	$Z_{in}(base) = \beta(r_e + r_e')$	For CC (emitter follower)
Stage input impedance	$Z_{in}(stage) = R_1 R_2 Z_{in}(base)$	Parallel combination
AC collector resistance	$r_c = R_C R_L$	Effective load at collector
Voltage gain (CE)	$A_v = r_c / r_e'$	Higher $I_E \rightarrow$ lower $r_e' \rightarrow$ higher gain
Voltage gain (CC)	$A_v = r_e / (r_e + r_e') \approx 1$	Always < 1 , non-inverting
Input voltage with loading	$v_{in} = Z_{in}(stage) / (R_G + Z_{in}(stage)) \times v_g$	Voltage divider with source resistance
Output voltage	$v_{out} = A_v \times v_{in}$	Use v_{in} (not v_g) for actual output
Output impedance CC	$Z_{out} = R_E (r_e' + (R_G R_1 R_2) / \beta)$	Very low output Z

15.4 H Parameters vs R Parameters

H Parameter	Equivalent R Parameter	Used For
h_{fe}	β (AC current gain)	Current gain in CE config
h_{ie}	$\beta \times r_e'$ (input impedance)	Input impedance at base
h_{re}	Negligible reverse voltage ratio	Usually ignored

H Parameter	Equivalent R Parameter	Used For
hoe	Negligible output conductance	Usually ignored

15.5 CE vs CC Amplifier Comparison

Feature	CE Amplifier	CC Amplifier
Output from	Collector	Emitter
Voltage gain	$r_c/r_e' (>> 1, \text{high})$	$r_e/(r_e+r_e') (< 1, \approx 1)$
Phase shift	180° (inverting)	0° (non-inverting)
Input Z	$\beta r_e'$ (moderate)	$\beta(r_e+r_e')$ (very high)
Output Z	$\approx R_C$ (high)	$r_e' + (R_G \parallel R_1 \parallel R_2)/\beta$ (very low)
Use case	Voltage amplification	Buffer, impedance matching

15.6 Common Exam Questions

13. Explain Voltage Divider Bias with circuit diagram. Derive step-by-step analysis procedure. (8 marks)
14. Explain Two Supply Emitter Bias. Draw circuit and derive analysis equations. (6 marks)
15. Compare different transistor biasing methods with respect to Q-point stability. (8 marks)
16. Explain the concept of small signal operation. What is the 10% rule? (4 marks)
17. Derive the AC emitter resistance formula $r_e' = 25\text{mV}/I_E$. (4 marks)
18. Explain T model and pi model of BJT transistor. Derive input impedance for each. (8 marks)
19. Derive the voltage gain expression $A_v = r_c/r_e'$ using both T and pi model. (6 marks)
20. Explain the loading effect of input impedance. Derive the expression for actual input voltage. (6 marks)
21. Explain CC amplifier (Emitter Follower). Derive voltage gain, input and output impedance. (8 marks)
22. Explain H parameters and their relation to R parameters (r_e' and β). (4 marks)
23. Write the procedure for DC analysis and AC analysis of a BJT amplifier. (6 marks)
24. Compare CE and CC amplifiers with respect to all performance parameters. (6 marks)

END OF MODULE 1 – ALL THE BEST FOR YOUR EXAMS! 🎓

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